

Data & Excel Analytics Interview Preparation

Comprehensive Solutions & Master Guide for Technical Interviews

Core Excel Concepts & Formulas

1. What is the difference between a workbook and a worksheet?

- **Workbook:** The entire Excel file (saved as `.xlsx`). It acts as a container that holds one or multiple worksheets.
- **Worksheet:** A single tab or spreadsheet page within a workbook consisting of a grid of cells (rows and columns) where data is entered and calculated.

2. Explain relative vs absolute cell references with an example.

- **Relative Reference (e.g., `A1`):** Changes automatically when you copy or drag the formula across cells. If you copy `=A1+B1` down one row, it updates to `=A2+B2`.
- **Absolute Reference (e.g., `$A$1`):** Fixed reference anchored using dollar signs (\$). It stays locked to cell A1 no matter where the formula is copied.
- *Example:* To calculate 18% tax on prices in column A: Formula in B2 is `=A2*$C$1` (where C1 contains the tax rate 0.18). Dragging B2 down keeps `$C$1` fixed while A2 changes to A3, A4, etc.

3. How does VLOOKUP compare to XLOOKUP?

Feature	VLOOKUP	XLOOKUP
Direction	Looks right only (lookup column must be leftmost)	Looks left, right, up, and down
Column reference	Requires manual column index number	Uses direct range references (safer against column additions)
Default match	Approximate match by default	Exact match by default
Error handling	Requires wrapping in <code>IFERROR()</code>	Has built-in <code>[if_not_found]</code> argument

4. Compare INDEX-MATCH versus XLOOKUP.

- **INDEX-MATCH:** The traditional power combination (`=INDEX(return_range, MATCH(lookup_value, lookup_range, 0))`). Highly flexible, fast, and compatible with older Excel versions.
- **XLOOKUP:** Modern replacement (`=XLOOKUP(lookup_value, lookup_range, return_range)`). Performs the exact same multi-directional lookup with cleaner, concise syntax and less likelihood of formula syntax errors.

5. When do you use SUMIF vs SUMIFS?

- **SUMIF:** Used when summing values based on a **single condition**. Syntax: `=SUMIF(range, criteria, [sum_range])`.
- **SUMIFS:** Used when summing values based on **multiple conditions**. Syntax: `=SUMIFS(sum_range, criteria_range1, criteria1, ...)`. Best practice is to default to `SUMIFS` even for single criteria as its argument order is more logical.

6. How does COUNT differ from COUNTA?

- **COUNT:** Counts cells containing **numeric values only** (numbers, dates). Ignores text, errors, and blank cells.
- **COUNTA:** Counts all **non-empty cells** (numbers, text, logical values, error codes).

Data Cleaning & Transformation

7. What is your workflow to remove duplicate records and handle missing values?

- **Handling Duplicates:** First, inspect unique identifiers. Use *Remove Duplicates* under the Data tab for quick removal, or use Power Query's *Remove Duplicates* step for dynamic, reproducible cleaning.
- **Handling Missing Values:**
 - Identify missing data using filters or `ISBLANK()`.
 - For numerical fields, impute missing values using mean/median or fill down/up if time-series.
 - For text/categorical fields, tag as "Unknown" or "N/A".
 - Drop incomplete records if missing values impair critical dimensions or analysis integrity.

Advanced Data Modeling & Power Tools

8. In what scenarios would you build a PivotTable?

- When you need to quickly aggregate large datasets without writing complex custom formulas.
- To summarize data dynamically across dimensions (e.g., Total Revenue by Region and Quarter).
- For ad-hoc exploratory analysis, slicing and dicing metrics, and building interactive dashboard summaries.

9. What is Power Query and why is it valuable for automated analysis?

- **Power Query:** An ETL (Extract, Transform, Load) engine built into Excel and Power BI.
- **Value:** It records every data cleaning step (unpivoting, filtering, merging, data type formatting) into M-code scripts. Upon receiving new raw data, clicking "Refresh" automatically re-executes all transformation steps, eliminating manual prep work.

10. Explain Merge vs Append operations in Power Query.

- **Merge (SQL Join equivalent):** Combines two tables horizontally based on a matching key column (e.g., joining Sales table with Customer details table).
- **Append (SQL UNION ALL equivalent):** Stacks two or more tables vertically, provided they have matching column names (e.g., combining Monthly sales files for Jan, Feb, and Mar).

11. What is a Data Model in Excel?

- A Data Model allows integration of data from multiple tables by establishing relational connections (1-to-many relationships) using primary and foreign keys.
- It enables building Star Schemas and running analyses on millions of rows using Power Pivot without flattening datasets into huge lookup-heavy sheets.

12. Define DAX and explain what a calculated measure is.

- **DAX (Data Analysis Expressions):** A formula language used in Power Pivot and Power BI for data modeling and custom calculations.
- **Calculated Measure:** Dynamic formulas that compute on the fly based on user filters/context in a dashboard or PivotTable (e.g., `Total Sales = SUM(Sales[Amount])`), as opposed to calculated columns which store static values in every row.

Dashboard Design & Auditing

13. How would you design an end-to-end sales performance dashboard?

- **Requirement Gathering:** Identify key business metrics (KPIs) like MoM Growth, Top Products, Regional Performance, and Conversion Rates.
- **ETL Pipeline:** Use Power Query to ingest, clean, and automate raw data feeds.
- **Data Modeling:** Load data into the Excel Data Model; define table relationships and create DAX measures for KPIs.
- **Visual Design:** Keep layout clean and modular (KPI cards on top, trends in the center, granular details at bottom). Use standard color palettes and interactive slicers.

14. How do you validate and audit your analytical models for accuracy?

- **Reconciliation:** Cross-check aggregated metrics against trusted raw source system totals or financial statements.
- **Data Audit Tools:** Use Excel's *Trace Precedents / Dependents* and *Formula Auditing* to verify logic paths.
- **Boundary & Stress Testing:** Test models with extreme values, zeroes, missing inputs, and unusual edge cases.
- **Automated Validation Checks:** Implement check-sum rows (e.g., `Source Total - Model Total = 0`) with conditional formatting flags.